

ENGINEERING DOCUMENTATION

YatraAI

Context-Aware Group Travel Intelligence Platform

A group itinerary planner in which a constraint solver — not a language model — decides what you do and when. Structured data owns the facts, deterministic algorithms select and schedule, a validator gates every plan, and the language model runs last and changes nothing. This document is the complete project reference and end-to-end technical workflow.

338 automated tests	130 curated places	−57% travel vs greedy	1.00 abstention accuracy	0 LLM calls required
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How to read this document. Sections 1–3 are what the system is and why. Section 4 is the complete technical workflow, stage by stage — it is the core of this document. Sections 5–11 are reference material: data, API, results, quality, security, operations and limitations.

1 · WHAT THIS PROJECT IS

YatraAI plans multi-day trips for **groups** across ten curated Indian destination circuits. A group states its dates, budget, pace, transport and accessibility needs; each member separately submits their own interests and constraints. The system returns a day-by-day schedule with real times, travel legs between stops, a cost range, weather notes, and a per-stop explanation of why that place was chosen for that group.

The interesting problem is not itinerary generation. It is producing a plan that is simultaneously **fair** across members with conflicting preferences, **explainable** at the level of individual decisions, and **physically feasible** under opening hours, travel time, budget and accessibility — without a language model inventing any of it.

1.1 The failure mode this design exists to prevent

Ask a general-purpose LLM to plan three days in Agra and it will confidently schedule the Taj Mahal on a Friday (closed to tourists), allow twenty minutes to cross Delhi, and quote a ticket price it invented. Each of those is a **constraint violation**, not a wording problem — so no amount of prompt engineering fixes it. The architecture separates the parts that must be correct from the part that must read well.

1.2 Feature inventory

AREA	CAPABILITY
Planning	Multi-day itinerary generation · per-day geographic clustering · CP-SAT scheduling · meal and rest breaks · per-day accommodation base with relocation notice
Group	Per-member preference submission · fairness-aware aggregation · member veto · opt-out · invite codes · change proposals with voting · trip chat
Knowledge	Cited Q&A; over a 790-chunk corpus · hybrid retrieval · abstention · computed citations · 'Know this place' detail drawer with history, etiquette, access and sources
Replanning	Eleven modification actions, each re-optimised and re-validated · weather-driven replanning · handling a member dropping out or a place becoming unavailable
Location	Consent-gated group location sharing · approximate-by-default · meeting point and ETAs · demonstration SOS that contacts nobody
Analytics	Feedback capture including actual spend · aggregate dashboard · admin data-quality, coverage, pipeline and evaluation views · provider telemetry

2 · RESPONSIBILITY SEPARATION

The system is eight stages. Seven are deterministic and testable. One is generative, and it runs last, after the plan is already chosen, scheduled and validated.

#	STAGE	OWNER	CAN IT INVENT FACTS?
1	Facts	Curated catalogue with per-row provenance	No — every field cites a source
2	Eligibility	Hard filters (closed, inaccessible, over budget, vetoed)	No — infeasible options are removed
3	Selection	Fairness-aware ranking, nine components	No — deterministic and reproducible
4	Geography	Balanced k-means day clustering	No
5	Travel	OSRM, falling back to haversine + documented speeds	No
6	Schedule	OR-Tools CP-SAT prize-collecting TSP with time windows	No — solves a stated model
7	Validation	Fifteen independent constraint checks	No — errors block display
8	Wording	LLM (optional)	Yes — which is why it runs last and changes nothing

Proof rather than assertion. The default LLM provider is mock. Live telemetry at `/api/v1/metrics` reports `llm.calls = 0`. Every itinerary produced during development and testing was generated with no model call at all. Configure a real provider and the plan is byte-identical — only the prose paragraph beside it changes.

3 · TECHNOLOGY STACK

LAYER	TECHNOLOGY	WHY THIS CHOICE
Frontend	Next.js 14 (App Router), TypeScript strict, Tailwind CSS, TanStack Query, Recharts, react-leaflet	Server components for static shells, client components only where state lives; strict TS catches API drift at build time
API	FastAPI, Pydantic v2, pydantic-settings, structlog	Schema-first validation and an OpenAPI document generated from the same types the code uses
Persistence	SQLAlchemy 2.0, Alembic, PostgreSQL + pgvector — portable to SQLite	Four portable TypeDecorators (GUID, JSONType, VectorType, TZDateTime) let one schema serve both engines, so a reviewer needs no infrastructure
Optimisation	Google OR-Tools CP-SAT	The problem is genuinely a prize-collecting TSP with time windows; AddCircuit expresses sequencing and selection in one constraint
ML / science	NumPy, pandas, scikit-learn, Pandera	Feature pipeline, offline experiments and executable data contracts
Data pipeline	Bronze/Silver/Gold medallion, Airflow DAGs, pyarrow	Idempotent, content-hashed ingestion with a contract at every promotion
Security	PyJWT (HS256), bcrypt, in-process sliding-window rate limiter	Token auth with a Supabase-compatible path; two rate-limit tiers protecting expensive endpoints
Testing / CI	pytest (295), Vitest + Testing Library (43), ruff, mypy, GitHub Actions	Backend, Postgres integration, data-quality, frontend and docker jobs
Deployment	Docker multi-stage, Compose, render.yaml, Vercel	Every secret marked sync:false; JWT_SECRET generated at deploy time

3.1 Provider abstraction

Every external dependency sits behind an interface with a documented fallback. No provider raises on failure — each degrades and records that it did, so the UI can disclose it and `/metrics` can report a real fallback rate.

PROVIDER	DEFAULT	FALLBACK	CONSEQUENCE OF FALLING BACK
LLM	mock	deterministic templates	Explanations become templated prose. The itinerary is unchanged.
Embeddings	hashing	— (is the default)	384-dim signed feature hashing. Deterministic, no model download; not a neural encoder.
Routing	OSRM	haversine × detour factor	Travel times use documented conservative speeds. Underestimating travel is the dangerous direction, so the fallback errs slow.
Weather	Open-Meteo	committed climatology	Labelled in the UI as seasonal averages, never presented as a forecast.
Database	PostgreSQL + pgvector	SQLite	Vector search degrades to in-process cosine. Same schema, same tests.

4 · COMPLETE TECHNICAL WORKFLOW

This section traces every significant path through the system end to end. It is the core of this document.

4.1 Workflow A — generating an itinerary

Triggered by POST `/api/v1/trips/{id}/itinerary`. Rate-limited on the AI tier (15 requests/minute).

STAGE	INPUT	PROCESS	OUTPUT
0 Authorise	JWT, trip id	Decode token, load trip, assert caller is a trip member. Rate-limit key is the user id.	TripContext + member list
1 Build context	Trip row, member preference rows	Merge trip-level settings with each member's interests, pace, mobility, day window, must-visit and avoid lists. Accessibility needs propagate up to the trip.	TripContext, MemberProfile[]
2 Load candidates	Cluster slug	Fetch every active attraction in the destination with schedules, costs and sources eagerly loaded; convert to pure-Python AttractionCandidate objects.	AttractionCandidate[]
3 Filter eligibility	Candidates, members	Remove anything closed for the whole trip, over budget, failing a stated accessibility need, or on any member's avoid list. A member veto is absolute.	Eligible candidates (reduced set)
4 Score & aggregate	Eligible candidates, members	Compute nine components per candidate per member; aggregate with <code>select_fairly</code> — an iterative selector that repeatedly picks the item maximising a blend of group objective gain and raw item quality.	ScoredAttraction[] with stored breakdowns
5 Cluster by day	Shortlist, trip length	Balanced k-means on an equirectangular projection with deterministic farthest-point seeding. A balancing move is refused if it would strand a point more than 25 km from where it belongs.	One candidate pool per day
6 Build travel matrix	Day pool + day base	Per-day base chosen within that day's area; if it sits more than 60 km from the stated origin the day carries an explicit relocation note. OSRM if available, else haversine × detour factor.	RouteMatrix (minutes, km)
7 Solve each day	DayPlanRequest	One CP-SAT model per day: AddCircuit over all nodes, time propagation via <code>OnlyEnforceIf</code> , opening-hour domains, mandatory nodes, activity/travel/budget caps. Five-step relaxation ladder on infeasibility.	PlannedDay with activities, status, objective, solve time
8 Validate	Planned days	Fifteen checks re-derived independently of the solver. Errors block display; warnings are surfaced.	ValidationReport
9 Cost & explain	Validated plan	Cost estimated as a range across five components with stated assumptions. The LLM (if configured) writes a summary paragraph; otherwise a deterministic template does.	Cost band, summary text
10 Persist	Everything above	New itinerary version written with per-day and per-activity rows, the stored score breakdown, solver stats and an input fingerprint. Previous version is superseded, not deleted.	Itinerary v(n), audit log entry

Why the fingerprint matters. Stage 10 stores a hash of the planner inputs. With `num_workers = 1` and a fixed seed, identical inputs always produce an identical plan — which is what makes the itinerary reproducible and the regression tests meaningful.

4.2 Workflow B — answering a question (RAG)

Triggered by POST `/api/v1/assistant/ask`. Also available unauthenticated, so the assistant can be demonstrated without signing up.

STEP	PROCESS
1 Sanitise	Detect prompt-injection patterns in the question; flag them in warnings and neutralise the text before it is used anywhere.
2 Classify	Map the question to a knowledge topic (history, significance, visiting, etiquette, access) using stem-based patterns.
3 Scope	Apply metadata filters. An attraction filter requires a cluster — an unscoped attraction lookup is exactly how the wrong place's dress code gets returned.
4 Retrieve twice	Dense cosine over the chunk embeddings and BM25 lexical scoring run independently over the same candidate set.
5 Fuse	Reciprocal Rank Fusion with <code>k = 60</code> combines the two rankings.
6 Rerank	A feature model scores query-term coverage, heading match, topic agreement and source authority.

STEP	PROCESS
7 Gate	Compute an absolute evidence score: $0.35 \times \text{dense similarity} + 0.65 \times \text{IDF-weighted query-term coverage}$, taken as a max over chunks. Below 0.26, abstain — with no model call at all.
8 Generate	If a real provider is configured, answer from neutralised context under an injection-aware system prompt. Otherwise compose from retrieved sentences.
9 Cite	Match each answer sentence back to the chunk it overlaps most (threshold 0.30). A sentence the model invented earns no citation.
10 Disclose	Attach the 'recorded, unverified' caveat for anything time-sensitive, and state explicitly when the question asked for a live value we do not hold.

4.3 Workflow C — replanning

Eleven actions are available: `remove_activity`, `replace_activity`, `regenerate_day`, `make_day_relaxed`, `reduce_cost`, `reduce_travel`, `add_theme`, `shift_start_time`, `weather_replan`, `member_opted_out`, `attraction_unavailable`.

Every one is expressed as a **change to planner inputs**, after which the entire pipeline in Workflow A re-runs and re-validates. Nothing edits a stored schedule in place. That is a deliberate constraint: an in-place edit could produce a state the optimiser would never have generated and the validator was never asked about.

On a trip with more than one active member the change does not apply immediately. It becomes a `ChangeProposal` with an approval threshold; the API returns `requires_group_approval: true` and a proposal id. Members vote, reaching the threshold auto-applies it, and the trip owner can override — with the override written to the append-only audit log.

4.4 Workflow D — data ingestion

LAYER	WHAT HAPPENS	MEASURED
Bronze	Seed JSON read and content-hashed. Re-running with unchanged content is a no-op, which is what makes the pipeline idempotent.	140 documents
Silver	Type coercion, deduplication, referential resolution, source allow-list enforcement, opening-window normalisation to minutes-from-midnight. Seven Pandera contracts gate the promotion.	130 attractions, 142 sources, 164 schedules, 216 fee rows, 0 rejected
Gold	Derived planning features: accessibility score, family score, indoor score, crowd penalty, cost index, evidence score, twelve monthly season scores.	130 feature rows, 10 cluster metric rows
Knowledge	The same records generate the RAG corpus — one chunk per attraction × topic — then embeddings. There is no second, driftable corpus.	140 documents → 790 chunks → 790 embeddings

4.5 Workflow E — location sharing

Consent text is served from the API so the UI cannot drift from the recorded version. A session cannot exist without `consent_granted_at` and a stored consent-text version. In approximate mode — the default — coordinates are snapped to a ~500 m grid **before** the row is constructed, so the precise value never reaches the database. Sessions carry a twelve-hour hard cap, points carry their own expiry, and stopping deletes the trail immediately rather than hiding it. Purging runs from three independent places so no single failure leaves data behind.

5 · DATA MODEL

Thirty-one tables across six domains, defined once in SQLAlchemy and migrated with Alembic. The full chain applies and reverses cleanly on a fresh database.

DOMAIN	KEY TABLES	NOTES
Catalogue	destination_clusters, attractions, attraction_schedules, attraction_costs, attraction_sources	Provenance is a first-class table, not a column. Every schedule and cost row carries verified = false.
Trips	trips, trip_members, member_preferences, change_proposals, votes, chat_messages	Preferences are per member, not per trip — that is what makes fairness measurable.
Itineraries	itineraries, itinerary_days, itinerary_activities, recommendation_scores	Versioned. Activities store the score breakdown and the opening window the visit was scheduled inside.
Knowledge	knowledge_documents, document_chunks, chunk_embeddings	Derived from the catalogue, so it cannot drift from it.
Privacy	location_sharing_sessions, location_points	Both carry expiry. There is no location-history table, by design.
Observability	audit_logs, analytics_events, model_runs, pipeline_runs, rag_evaluations	model_runs carries data_kind on every row, so a synthetic result can never be displayed as a real one.

6 · API SURFACE

GROUP	ENDPOINTS	AUTH
Ops	/health · /ready · /api/v1/metrics · /auth/service-status	public
Auth	register · login · me · demo	public / bearer
Destinations	list · detail · attractions · attraction detail	public
Trips	create · list · detail · update · join · invite rotate · preferences · opt-out	member / owner
Itineraries	generate · read · versions · validate · modify · recommendations	member
Assistant	ask · suggested-questions · retrieve · why	public / member
Collaboration	votes · proposals · apply · chat	member / owner
Location	consent-text · start · status · point · group · delete	member (consent-gated)
SOS	raise · list · resolve	member (demo only)
Analytics / Admin	feedback · dashboard · trip analytics · data-quality · coverage · pipeline-runs · rag-evaluations · audit-log	public / admin

Every error returns the same envelope so clients branch on a stable code rather than parsing messages: {"error": {"code", "message", "detail"}}. Notable codes include planning_infeasible (422, with the report attached), consent_required (403) and rate_limited (429, with retry-after).

7 · MEASURED RESULTS

Reproduce with `make experiments` and `python evaluation/run_rag_eval.py`.

7.1 Optimiser versus a fair greedy baseline

METRIC	GREEDY	CP-SAT	DIFFERENCE
Total travel distance	149.8 km	64.0 km	−57%
Total travel time	487 min	275 min	−43%
Travel km per activity	15.61	6.01	−61%
Activities scheduled	9.8	10.3	+0.5
Constraint violations	0.0	0.0	—
Median computation	8.8 ms	48.2 ms	+39 ms

Six scenarios across six clusters, three timing repeats. The baseline honours identical opening hours, budget, activity and travel caps, so the gap measures sequencing quality alone — it is not a strawman.

7.2 Group aggregation methods

METHOD	MEAN SATISFACTION	LEAST SATISFIED	SPREAD
simple_average	0.604	0.481	0.172
borda_count	0.618	0.478	0.197
max_min_fairness	0.582	0.545	0.069
fairness_aware (single pass)	0.603	0.484	0.167
select_fairly (production)	0.597	0.522	0.115

Five group profiles built so a simple average steamrolls a minority. Note that Borda scores the best mean and the worst floor — that is the trap this measurement exists to expose. The single-pass variant barely improves on an average, which is precisely why the iterative selector was built.

7.3 Retrieval and abstention

METRIC	VALUE	METRIC	VALUE
Open retrieval top-1	0.903	Citation correctness	1.000
Open retrieval top-3	1.000	Citation coverage	1.000
Open retrieval MRR	0.952	Abstention accuracy	1.000
Retrieval recall	1.000	Faithfulness	0.995
Topic classification	0.941	Mean evidence (answerable / out-of-scope)	0.582 / 0.071

Thirty-four hand-written questions including deliberately unanswerable and adversarial ones. Ablation, with a retrieval arm genuinely disabled rather than re-sorted: reranking is worth +12.9 points of top-1 (0.903 against 0.774). Lexical-only nearly matches the hybrid, which is an honest limitation of the default hashing embedder — a lexical projection whose signal correlates with BM25 rather than complementing it.

7.4 The machine-learning experiment

A suitability model was trained on **synthetic labels** — there is no genuine labelled data for "did this group enjoy this attraction". Every run is stamped `data_kind = "synthetic"`.

MODEL	ROC-AUC	PR-AUC	NOTE
logistic_regression	0.6711	0.8341	best learner
random_forest	0.6404	0.8190	lost to the control
gradient_boosting	0.6404	0.8147	lost to the control
rule_based_production_control	0.6676	0.8278	the shipped scorer

The honest result. The tree models lost to the hand-weighted production scorer. The only learner that edged it did so by 0.0035 ROC-AUC, and a paired bootstrap over 2,000 resamples puts that difference at **95% CI [-0.0100, +0.0173]** — straddling zero. On this data the simple transparent model is not a compromise. No model was promoted to production.

8 · TESTING & QUALITY

SUITE	COUNT	WHAT IT COVERS
Unit	part of 295	Scoring, aggregation, fairness metrics, clustering, cost estimation, validator, retrieval scoring, security primitives
Integration	part of 295	Full API surface against a live app and database — auth, trips, itineraries, RAG, collaboration, location, analytics, admin
Data quality	30	Coordinate bounds, cluster distance, duplicate detection, category-taxonomy coverage, source allow-list, dress code on religious sites, verified-flag enforcement
Frontend	43	Design-system accessibility semantics, itinerary rendering, opening-window display states, solver-trace parsing including malformed input
Total	338	ruff and ruff format clean; mypy clean; tsc --noEmit clean; next lint clean

8.1 Defects found by measurement, not by reading code

DEFECT	HOW IT WAS FOUND	FIX
Topic classification stuck at 0.50	RAG evaluation harness	A trailing word boundary in <code>\b(histor)\b</code> makes matching "history" impossible. Switched to <code>\w*</code> stems: 0.50 → 0.941.
Abstention could not separate answerable from unanswerable	RAG evaluation harness	The threshold keyed on the fused RRF score, which encodes ordering only and is near-constant at the top. Replaced with an absolute evidence score.
Assistant abstained on a fee it was holding	RAG evaluation harness	"today" and "4pm" are absent from the corpus and drew maximum IDF weight. Absent entities signal out-of-scope; absent framing words signal nothing. Abstention 0.912 → 1.000, with out-of-scope evidence unchanged at 0.071.
A 401 km Delhi–Agra itinerary	Planner comparison experiment	Size balancing dragged Delhi sites into the Agra day. Added a rule refusing any balancing move that strands a point more than 25 km away. Same scenario now plans at 54 km.
Every Delhi–Agra day infeasible	Planner comparison experiment	A single trip-wide base sat 115 km from everything. Introduced a per-day base with an explicit relocation note.
Cross-cluster neighbour reference; duplicate coordinates	Data-quality contracts	Two authoring errors in the seed data, neither visible by reading the JSON.

9 · SECURITY & PRIVACY

CONCERN	CONTROL
Credential compromise	bcrypt at cost 12 with a SHA-256 pre-hash, because bcrypt silently truncates at 72 bytes. A test asserts an 80-character passphrase keeps its entropy.
Token forgery	HS256 JWT with signature, issuer and expiry verified. Supabase-issued tokens are accepted and provision a local user on first sight.
Privilege escalation	require_trip_member and require_trip_owner as route dependencies; admin_user on every admin route. Admins may read a trip for support but are not silently made members.
Cost and denial of service	Two-tier sliding-window rate limiting — 120 requests/minute default, 15/minute on the optimiser and the assistant.
Prompt injection	Detected on the question, neutralised on every retrieved chunk, and an injection-aware system prompt. Injection attempts also score low on evidence and usually abstain before any model call.
Data poisoning	Source-domain allow-list enforced at the Silver gate; a citation to a non-allow-listed domain is rejected with a reason.
Secret leakage	.env is git-ignored, .env.example contains variable names only, render.yaml marks every secret sync:false, and a production boot guard refuses to start on weak configuration.
Location exposure	Consent gate, ~500 m snapping before storage, hard expiry, immediate deletion on stop, and coordinate-key stripping at analytics write time.

10 · RUNNING THE PROJECT

One command on Windows:

```
.\start.ps1
```

Creates the virtual environment, installs, migrates, seeds, starts the API and opens a browser.

Full stack, manually:

```
python -m venv .venv
.venv\Scripts\pip install -e ".[dev]"
.venv\Scripts\python -m yatraai.cli migrate
.venv\Scripts\python -m yatraai.cli seed --knowledge --embeddings
.venv\Scripts\uvicorn yatraai.main:app --app-dir apps/api --port 8000

cd apps/web && npm install && npm run dev
```

Frontend at localhost:3000; the API's self-contained console at localhost:8000 (it renders offline — /docs needs internet because Swagger UI loads from a CDN). Demo account: demo@yatraai.example / yatraai-demo-2026, or one click on the sign-in page.

10.1 Repository layout

PATH	CONTENTS
apps/api/yatraai/	FastAPI application — api/, services/ (planner, recommend, rag, routing, weather, llm, location, analytics), db/, schemas/, seed/
apps/web/	Next.js frontend — app/ routes, components/, lib/ (typed API client and types)
data/seed/attractions/	Ten curated destination files; the committed source of truth
pipelines/	Medallion transforms and Airflow DAGs calling the same functions as the CLI
ml/	Experiments and reports — planner comparison, aggregation comparison, suitability model
evaluation/	RAG benchmark dataset, harness and generated report
tests/	unit/, integration/, data_quality/
docs/	Architecture, data flow, ER diagram, recommendation, optimisation, RAG, ML experiments, security, privacy, deployment, API reference, data dictionary, interview notes
scripts/	Deployment verification, deck and document generators

11 · STATED LIMITATIONS

These are deliberate scope decisions or honest gaps. They are listed here rather than discovered later.

LIMITATION	DETAIL
Default embeddings are not neural	A 384-dimension signed feature hashing projection. It keeps installation under a minute and requires no model download. sentence-transformers is a one-environment-variable upgrade, and the ablation table shows what retrieval configuration is worth on this corpus.
No model was promoted to production	The ML experiment runs on synthetic labels and lost to the rule-based control. The pipeline, registry and evaluation are real and ready for the day genuine feedback exists.
The benchmark is author-written	Thirty-four questions, hand-written, not collected from users. Small, and labelled as such in the report.
Ten destinations, not all of India	Depth over breadth. Adding a cluster is a data change plus climatology, with no application code change.
Rate limiting is per-process	Deliberate for a single-process demo. SlidingWindowRateLimiter.check is the only method the API uses, so a distributed implementation is a one-class swap.
No refresh tokens, no 2FA, no account deletion endpoint	Access tokens last 120 minutes. Cascades are defined at the schema level but the endpoint is not built.
Weather and routing run on fallbacks offline	Both are disclosed in the UI and reported in /metrics rather than hidden.
SOS is a demonstration	It contacts nobody. Stated in the OpenAPI description, the response body, the UI and the chat message it posts, and it refuses to fire without an explicit acknowledgement flag.

Honesty guarantees, enforced by code rather than by policy. No schedule or fee row in the dataset claims to be verified, and a data-quality test fails the build if one ever does. The assistant abstains rather than guessing. Citations are computed from lexical overlap, so a model cannot claim a source it did not draw from. Coordinates never reach analytics. Every performance figure in this document is regenerated by a script in the repository.